

REUS AP, Spain

WMO: 081750

Lat: 41.1497N

Lon: 1.1789E

Elev: 249

StdP: 14.56

Time Zone: 1.00 (EUC)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	30.5	33.6	12.5	10.5	43.3	18.0	13.8	40.8	32.4	54.0	29.9	54.8	4.3	20	0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	17.0	90.3	71.8	88.1	71.7	86.3	71.5	76.7	84.5	75.5	83.6	74.4	82.6	10.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.5	130.2	81.3	73.1	124.0	80.6	71.7	118.2	80.0	40.3	84.6	39.1	84.0	38.1	82.6	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.2	20.3	17.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	62.0	48.0	49.5	53.4	58.1	64.3	72.1	77.7	78.0	71.8	65.2	55.0	50.3
	DBStd	11.56	5.67	5.37	4.32	3.88	4.34	4.48	3.21	3.30	4.33	4.97	5.94	5.01
	HDD50	264	102	65	17	1	0	0	0	0	0	0	22	57
	HDD65	2420	527	434	360	210	65	3	0	0	6	57	301	457
	CDD50	4651	40	51	121	243	444	663	860	867	654	471	172	65
	CDD65	1332	0	0	0	2	44	215	395	402	209	64	1	0
	CDH74	10598	0	0	4	20	244	1622	3488	3576	1335	307	2	0
	CDH80	3241	0	0	0	0	19	449	1229	1242	271	31	0	0
Wind	WSAvg	6.6	7.6	7.5	7.3	6.8	6.7	6.1	6.3	5.9	6.1	5.8	7.2	6.5
Precipitation	PrecAvg	22.1	1.4	1.2	1.5	1.7	2.1	1.7	0.8	2.0	2.6	3.4	2.2	1.6
	PrecMax	37.8	5.7	3.9	7.3	8.5	4.9	7.5	3.6	10.6	10.5	10.9	8.5	5.8
	PrecMin	13.3	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0
	PrecStd	6.5	1.2	0.9	1.4	1.7	1.3	1.8	1.0	2.1	2.4	2.8	2.0	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	69.2	70.3	72.3	76.6	82.1	92.3	92.2	95.2	88.2	84.0	72.7	66.6
		MCWB	56.6	55.9	55.9	60.2	63.5	71.1	71.1	72.2	70.7	62.9	61.4	56.6
	2%	DB	64.7	65.8	69.0	72.7	79.4	87.6	89.9	90.5	85.0	79.2	70.0	64.4
		MCWB	53.3	53.3	54.6	58.5	62.4	68.8	72.0	72.5	69.9	65.2	60.6	54.6
	5%	DB	62.0	62.5	66.3	70.0	76.8	84.9	88.0	88.0	83.0	77.3	67.8	62.2
		MCWB	52.1	51.3	53.6	57.9	62.3	68.9	72.5	72.0	69.4	65.6	59.2	53.3
	10%	DB	58.8	60.0	63.8	67.6	73.9	82.2	86.3	86.1	80.9	75.1	65.0	59.9
		MCWB	50.1	50.2	52.5	56.7	61.2	67.7	72.1	72.0	68.3	65.0	56.7	52.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.2	57.0	59.1	62.1	67.2	74.8	78.0	78.9	75.6	70.9	65.3	58.5
		MCDB	66.8	66.2	67.8	71.2	75.5	85.7	85.5	86.2	83.4	77.0	69.3	63.4
	2%	WB	55.0	55.1	56.9	60.6	65.6	73.0	76.4	77.3	74.0	69.3	63.0	56.4
		MCDB	62.3	62.7	65.3	69.3	74.4	83.7	84.7	84.6	81.1	75.2	67.7	61.7
	5%	WB	53.1	53.3	55.4	59.5	64.3	71.2	75.4	76.0	72.6	68.0	60.8	55.0
		MCDB	60.1	60.3	62.9	67.1	72.3	80.5	83.9	83.5	79.4	73.4	65.6	59.9
	10%	WB	51.3	51.6	54.0	58.2	63.1	69.5	74.3	74.9	71.3	66.7	58.5	53.2
		MCDB	57.6	57.4	61.2	65.0	71.0	78.8	82.7	82.6	78.0	71.8	63.2	58.8
Mean Daily Temperature Range	5% DB	MDBR	18.3	18.7	19.1	17.5	18.5	18.1	16.6	17.0	17.7	17.7	18.1	18.1
		MCDBR	21.5	23.1	24.3	21.9	21.7	20.9	19.2	19.5	19.5	20.4	19.6	20.4
		MCWBR	13.4	13.7	13.6	11.2	10.1	9.5	9.2	9.8	9.7	10.7	11.6	12.9
	5% WB	MCDBR	19.2	20.7	20.5	18.5	17.5	18.5	16.5	16.0	17.0	16.5	15.7	18.9
MCWBR		12.8	13.1	12.8	10.5	9.1	9.0	8.7	8.6	9.1	9.9	10.5	12.6	
Clear-Sky Solar Irradiance	taub	0.313	0.333	0.378	0.397	0.418	0.433	0.437	0.437	0.414	0.385	0.339	0.314	
	taud	2.518	2.464	2.351	2.316	2.295	2.292	2.298	2.298	2.356	2.427	2.498	2.519	
	Ebn at Noon	269	279	276	278	273	268	266	262	261	256	255	258	
	Edh at Noon	23	28	36	39	41	41	41	40	35	29	24	22	
All-Sky Solar Radiation	RadAvg	666	963	1388	1737	2063	2302	2292	1970	1520	1074	726	577	
	RadStd	56	106	109	117	138	74	74	74	74	82	69	44	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+193	+151

Nomenclature: See separate page